# SceneSplat++: A Large Dataset and Comprehensive Benchmark for Language Gaussian Splatting

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NEURAL INFORMATION PROCESSING SYSTEMS

## SceneSplat Series

ShapeSplat [3DV'25 Oral] → SceneSplat [ICCV'25 Oral] → **SceneSplat++ [Neurips'25]** → Chorus [arXiv]

## Motivation

**Towards efficient open-vocabulary interaction within Language GS.**

*Segmentation*
*Text-Query*
*Scene Editing*
*...*

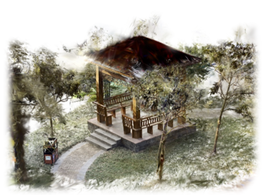
⚠ **Current Eval Limitations**

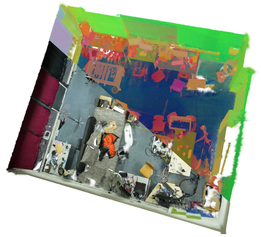
- **Small number of test scenes** → **Large 3DGS Dataset**
- **Tested close to training views** → **Comprehensive Benchmark**
- **Only 2D metrics** → **3D&2D Metrics**

🎯 **Our Solutions**

## SceneSplat-49K Dataset

- **48,856** Indoor/Outdoor Scenes
- **12,061** With Language Embedding
- **26.12** Billion 3DGS in Total
- **~ 2.36** L4 GPU Years

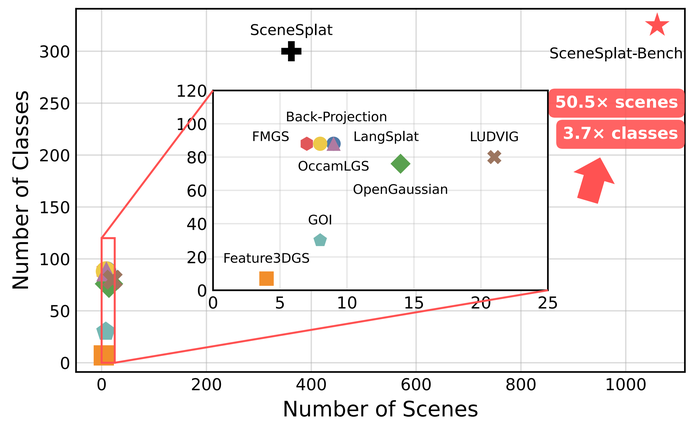


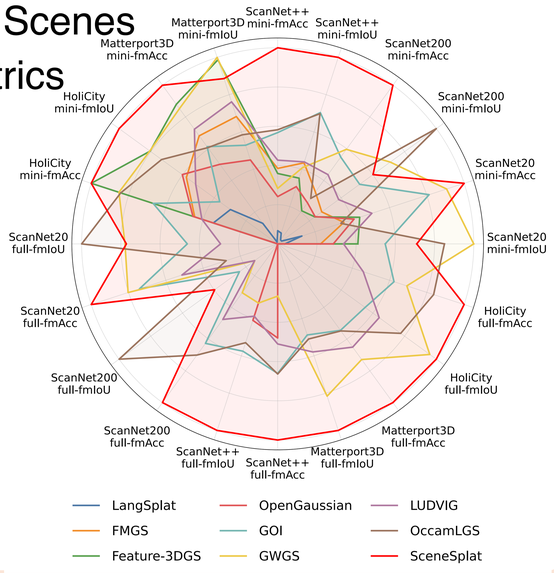


- Per-3DGS Language Embedding
- Diverse Scene(1.7 $m^2$ ~ $2.1\times10^6 m^2$)
- High Quality (Avg. PSNR 27.83)
- RGB Frames 22.65 Million

## SceneSplat-Benchmark

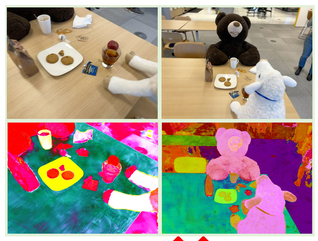
- **3D** Eval Protocol • **5** Datasets • **1076** Scenes
- **325** Classes • **9** LGS Methods • **6** Metrics




## Key Findings & Insights

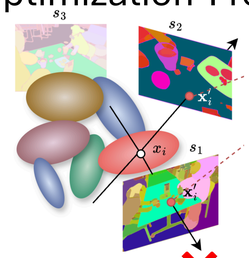
### 1. Per-Scene Optimization Methods



Scalable ❌
Efficiency ❌ ~ 300min

*e.g., LangSplat (Minghan et al., 2024)*

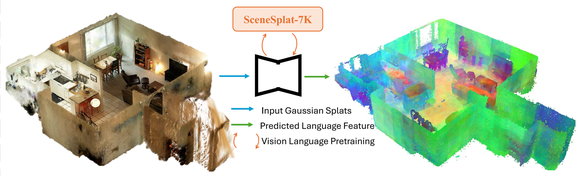
### 2. Per-Scene Optimization-Free Methods




Scalable ❌
Efficiency ✅ ~ 5min

*e.g., OccamLGS (Jiahuan et al., 2024)*

### 3. Generalizable Method




Scalable ✅
Efficiency ✅ ⚡ ~ 14s

*e.g., SceneSplat (Yue et al., 2025)*

- Benchmark Result: Group1< Group2<< Group3
- Generalizable methods outperforms per-scene methods. Pros: Scalable, Efficiency.
- Scaling up the generalizable method on SceneSplat-49K improves performance.